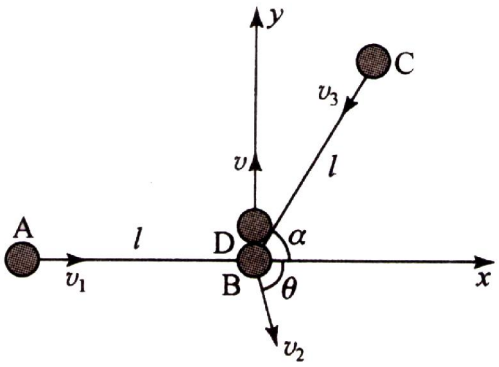
24th CPhO Final – Solutions

1.

1. Post-impact directions: elastic head-on impact of D and B leaves D moving along y, toward positive y.

with speed v. In a collision of D , B the strings give A , C impulses. The strings are flexible, so A , C feel no transverse force. Thus then, A moves along AB, and C along CB. θ is the angle of B's velocity to x.



The directions are shown. The two strings linking A , B , C break at once, so each ball subsequently moves at constant velocity.

2. Post-impact speeds: v_1 , v_2 , v_3 are speeds of A , B , C respectively, as drawn. Momentum is conserved, so along x we have

$$mv_1 - mv_3\cos\alpha + mv_2\cos\theta = 0 \ ; \qquad (1)$$

Along y we have

$$-mv_0 = mv - mv_2\sin\theta - mv_3\sin\alpha \ . \qquad (2)$$

Conservation of energy gives

$$\frac{1}{2}mv_0^2 = \frac{1}{2}mv_1^2 + \frac{1}{2}mv_2^2 + \frac{1}{2}mv_3^2 + \frac{1}{2}mv^2 \ . \qquad (3)$$

Inextensible strings: B , C: equal BC-velocity components; A , B: equal AB-velocity components. Hence

$$v_2\cos\theta = v_1 \ , \qquad (4)$$

$$v_2\cos[\pi - (\alpha+\theta)] = v_3 \ . \qquad (5)$$

Substituting $\alpha = \pi/3$ and solving the equations gives

$$v_1 = \frac{\sqrt{3}}{12}v_0 \ , \qquad (6)$$

$$v_2 = \frac{\sqrt{21}}{6}v_0 \ , \qquad (7)$$

$$v_3 = \frac{\sqrt{3}}{3}v_0 \ , \qquad (8)$$